

Outline Draft

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1 Introduction

Alzheimer's Disease (AD) is characterized as a complex neurodegenerative disease affecting more than 55 million people globally [12]. Symptoms are caused by the degeneration of the central nervous system, and are manifested by progressive memory loss, difficulty with language, drastic mood swings, among other effects. It is a disease that causes an extensive burden to those suffering from the disease through disorientation, to relatives in the form of mental taxation, as well as economically, projecting an estimated global economic burden of 16.9 trillion dollars by 2050 [16]. These factors are just a few reasons as to the importance of research in this area, where many scientists have worked to understand the underlying mechanisms of AD. However, much is still unknown in how AD is caused, as well as what biological processes occur in the brain, making it difficult to come to a decisive conclusion regarding its cause.

Many studies have been conducted in regard to understanding Alzheimer's Disease. With this, there are a few main hypotheses that are generally considered to be factors in the causation and progression of AD [20]. The so-called *Amyloid Cascade Hypothesis* is among the most extensively studied in the literature, where it is proposed that a misfolding of β -amyloid (A_β) protein forms senile plaques. These plaques are hypothesized to be the main driving force in the intracellular deposition of misfolded τ proteins in neurofibrillary tangles (NFTs) and memory loss in the progression of Alzheimer's Disease [7]. Growing evidence, however, shows that this hypothesis is suggested to be but a small factor in the larger picture of the underlying operative mechanisms of the disease [10]. This has led researchers to branch into other pathways concerning AD, looking at the interactions between A_β and τ in an attempt to further understand the pathogenesis of these two factors [19]. Some studies have also observed the role of calcium in this process, further suggesting a more complex process that is occurring biologically. This occurs through a feedback interaction between an upregulation of Ca^{2+} and the amyloidogenic pathway, remodeling the Ca^{2+} signaling pathway [4].

The various complex interactions in AD have shown to be an obstacle to yet overcome when modeling the behavior of the disease. Numerous different studies have in turn given an attempt at creating mathematical models observing different aspects of the disease in order to observe the underlying dynamics. This has led to many models being introduced, each aiming to understand various components in Alzheimer's Disease through different methodologies. Compartments concerning the interaction between A_β and τ are studied, observing the suspected effects on neurodegeneration and cognitive decline through a mechanistic causal model [18, 6]. The feedback loop between A_β and Ca^{2+} has also been analyzed using a

deterministic system, adding a noise-inducing element to the system [17, 21]. Models that observe lower-level processes have also been studied, lending to valuable insight as to what is suggested to be seen on the cellular level [13, 11].

It should be noted that in these models, while each has its own way of observing some of the underlying mechanisms, there are still potential shortfalls when it comes to finding effective treatments. Evidence suggests that focusing solely on anti- A_β treatments or τ protein treatments, rather than attempting to determine treatments for many different aspects of the disease [1, 19]. This provides difficulty in understanding exactly the importance of which component of the disease should be the focus of further study in order to provide evidence for treatments having higher efficacy. In this study, a mechanistic model is proposed, analyzing the interaction between A_β , Ca^{2+} , and τ proteins; observing the effects of this multifaceted interaction on neurodegeneration, and cognitive decline. Sensitivity analysis is performed in order to better understand the importance of each aspect of the model, and optimal control is used in order to determine the suggested avenues of effective treatment.

2 Proposed Model

To understand the role calcium may have in the interaction of AD, a model is proposed using five biomarkers found to have interactions within the disease: hallmark biomarkers through amyloid beta and tau, an interactional biomarker through calcium, suggested to have a role in amyloid beta and tau protein accumulation. Also observed are neuronal loss biomarkers, and cognitive decline. The pathology of AD is suggested to start through a feedback loop between amyloid beta and calcium through a dysregulation of calcium; this exacerbates the aggregation of amyloid beta, forming plaques. These plaques then promotes the hyperphosphorylation of tau proteins, leading to neurodegeneration and subsequently cognitive decline.

2.1 Amyloid Beta and Calcium Equations

Because amyloid beta and calcium are suggested to interact in such a way that a dysregulation of calcium causes an acceleration of amyloid beta accumulation [17], this is modeled mathematically as

$$\frac{dA_\beta}{dt} = a_1 + a_2 \frac{Ca}{Ca + \sigma} - k_1 A_\beta - u_1 A_\beta \quad \text{with} \quad A_\beta(T_0) = A_0, \quad (1)$$

where A_0 is the initial condition of A_β at age T_0 , and can vary on a per patient basis [6]. The rate of A_β growth increases at a linear rate a_1 and decreases proportionally to the population of A_β , as described by $k_1 A_\beta$. Here, $a_2 \frac{Ca}{Ca + \sigma}$ acts as the interaction of calcium on A_β using a Hill equation of $n = 1$. The removal rate can be modified by choosing different values for u_1 which represents the effect of treatment targeting A_β . The term u_1 can either be constant or be a function of time.

Similarly, an A_β interaction is observed to have a role in the further dysregulation of calcium, and the equation for calcium is defined as

$$\frac{dCa}{dt} = b_1 + b_2 A_\beta - k_2 Ca - u_2 Ca \quad \text{with} \quad Ca(T_0) = Ca_0. \quad (2)$$

Here, calcium grows at a linear rate b_1 , and the accumulation of A_β is positive at a rate b_2 . There is also a natural removal rate of calcium, k_2 . Notice that there is also the option to administer treatment targeting calcium through the term u_2 .

2.2 Tau Equation

Many studies have suggested that the pathology of tau is initiated by amyloid beta accumulation, and have modeled this process mathematically [22, 6]. However, it has also been suggested that calcium dysregulation also has an effect inducing the production of tau proteins [5]. This leads to a definition of the tau equation to be given by

$$\frac{d\tau}{dt} = c_1 + c_2 A_\beta + c_3 Ca - k_3 \tau - u_3 \tau \quad \text{with} \quad \tau(T_0) = \tau_0. \quad (3)$$

It is noted here that only pathological tau is observed, where other studies have used compartments relating to non-amyloid dependent tau accumulation, or suspected non-Alzheimer pathology (SNAP) [18, 6]. This function is similar to Equation (1) and Equation (2) in that $\frac{d\tau}{dt}$ grows at a linear rate d_1 and decreases proportional to the population of τ through $k_3 \tau$. There is also the treatment term $u_3 \tau$.

2.3 Neurodegeneration Equation

When tau is in a pathological state, it no longer binds to microtubules as intended; instead, the formation of NFTs occur in the neurons, leading to neurodegeneration [2]. Hence, this equation is defined as

$$\frac{dN}{dt} = d_1 + d_2 \tau - k_4 N.$$

This equation follows the established structure and also includes the interaction term $d_2 \tau$.

2.4 Cognitive Decline Equation

Both tau and neurodegeneration are evidenced to directly affect cognitive decline [3, 6]. Because of this, the equation for cognitive decline is defined as

$$\frac{dC}{dt} = e_1 + e_2 NR + e_3 \tau - k_5 C.$$

The parameter R represents a patient having the APOE $\epsilon 4$ gene, which has been shown to be a risk factor in Alzheimer's disease (need to find citation of this).

This model is summarized as a system of ordinary system of differential equations with

$$\begin{cases} \frac{dA_\beta}{dt} = a_1 + a_2 \frac{Ca}{Ca+\sigma} - k_1 A_\beta - u_1 A_\beta \\ \frac{dCa}{dt} = b_1 + b_2 A_\beta - k_2 Ca - u_2 Ca \\ \frac{d\tau}{dt} = c_1 + c_2 A_\beta + c_3 Ca - k_3 \tau - u_3 \tau \\ \frac{dN}{dt} = d_1 + d_2 \tau - k_4 N \\ \frac{dC}{dt} = e_1 + e_2 NR + e_3 \tau - k_5 C \end{cases} \quad \text{with} \quad \begin{cases} A_\beta(T_0) = A_0 \\ Ca(T_0) = Ca_0 \\ \tau(T_0) = \tau_0 \\ N(T_0) = N_0 \\ C(T_0) = C_0. \end{cases} \quad (4)$$

3 Analysis

3.1 Preliminary Results

Assuming Equation (4) is subject to initial conditions that are non-negative, it is noted that the system is invariant in the non-negative orthant. Let $r_1 = k_1 + u_1$ and $r_2 = k_2 + u_2$ for the simplicity of calculations. Considering the components uncoupled from the rest of the proposed model, $dA_\beta/dt, dCa/dt$ can be shown to be bounded. This is done by first noticing

$$a_1 + a_2 \frac{Ca}{Ca + \sigma} - r_1 A_\beta \leq a_1 + a_2 - r_1 A_\beta$$

since $Ca/(Ca + \sigma) \in [0, 1)$. So, $a_1 + a_2 - r_1 A_\beta < 0$ when $A_\beta > \frac{a_1 + a_2}{r_1}$. Similarly, using that A_β is bounded, it follows that

$$\frac{dCa}{dt} = b_1 + b_2 A_\beta - r_2 Ca \leq b_1 + b_2 M - r_2 Ca$$

as $M > A_\beta(t)$ for all $t \geq 0$. So, $b_1 + b_2 M - r_2 Ca < 0$ when $Ca > \frac{b_1 + b_2 M}{r_2}$, thus Ca is also bounded. The rest of the system follows from these two variables, so the rest of the system is bounded as well.

Lemma 1 (Establishment of Unique Equilibria of Uncoupled System). *There exists a unique equilibria (A_β^*, Ca^*) for the system*

$$\begin{cases} \frac{dA_\beta}{dt} = a_1 + a_2 \frac{Ca}{Ca + \sigma} - r_1 A_\beta \\ \frac{dCa}{dt} = b_1 + b_2 A_\beta - r_2 Ca. \end{cases} \quad (5)$$

Proof. To establish the existence of equilibria for Equation (5), a solution to the following must be obtained:

$$\begin{aligned} a_1 + a_2 \frac{Ca}{Ca + \sigma} - r_1 A_\beta &= 0 \\ b_1 + b_2 A_\beta - r_2 Ca &= 0. \end{aligned}$$

By solving $\frac{dA_\beta}{dt} = a_1 + \frac{a_2 Ca}{Ca + \sigma} - r_1 A_\beta$ for A_β , we get

$$A_\beta = \frac{a_1 Ca + a_2 Ca + a_1 \sigma}{r_1 (Ca + \sigma)}.$$

We then substitute this value into $\frac{dCa}{dt} = b_1 + b_2 A_\beta - r_2 Ca$. The resulting expression is,

$$b_1 - r_2 Ca + \frac{b_2(a_1 Ca + a_2 Ca + a_1 \sigma)}{r_1 (Ca + \sigma)}. \quad (6)$$

Setting Equation (6) equal to 0 and doing further simplification yields the quadratic,

$$-(r_1 r_2) Ca^2 + (a_1 b_2 + a_2 b_2 + b_1 r_1 - r_1 r_2 \sigma) Ca + a_1 b_2 \sigma + b_1 r_1 \sigma = 0. \quad (7)$$

Notice that in Equation (7) the quadratic term is always negative and the constant term is always positive. The linear term can either be positive or negative depending on the choice of parameters. Regardless of the parity of the linear term, there is only one sign change in the expression, so by Descartes' Rule of Signs, there is one positive real root to this quadratic. Thus, there exists a unique positive equilibrium (A_β^*, Ca^*) for the system in Equation (5). $\square$

Lemma 2 (Global Stability of Equilibrium). *The equilibrium (A_β^*, Ca^*) is globally asymptotically stable.*

Proof. Let $\mathbf{F}(A_\beta, Ca) = \langle a_1 + \frac{a_2 Ca}{Ca + \sigma} - r_1 A_\beta, b_1 + b_2 A_\beta - r_2 Ca \rangle$. Then, $\nabla \cdot \mathbf{F} = -r_1 - r_2$. Choose the function $\alpha(A_\beta, Ca) = 1$. So, $\nabla \cdot (\alpha \mathbf{F}) < 0$, therefore by the Bendixson-Dulac Theorem, there is no limit cycle in the system of ODEs. So, the equilibrium point must be globally stable in the non-negative orthant. $\square$

Theorem 1 (Global Stability of Equilibrium of System). *There exists a unique globally asymptotically stable equilibrium $(A_\beta^*, Ca^*, \tau^*, N^*, C^*)$.*

Proof. By Lemma 2, there exists a unique global asymptotically stable equilibrium (A_β^*, Ca^*) . By solving (3) for τ we get the following equilibrium value:

$$\tau^* = \frac{c_1 + c_2 A_\beta^* + c_3 Ca^*}{k_3 + u_3}.$$

By similar methods we find

$$N^* = \frac{d_1 + d_2 \tau^*}{k_4}$$

and

$$C^* = \frac{e_1 + e_2 N^* R + e_3 \tau^*}{k_5}.$$

Thus there is a unique global asymptotically stable equilibrium solution for the system in Equation (4). $\square$

4 Parameters

The values for the 19 key parameters in the causal model are obtained and estimated from the literature, and are summarized in Table 1. In order to provide a plausible model, parameters used in previous studies are adjusted to allow for similar units and orders of magnitude. When this is not possible, parameters are estimated using values found from various biological studies as are credible.

Table 1: Parameters used for simulation

Var	Value	Unit	Source
A_0	0	micromolars (μM)	-
Ca_0	100	nanomolars (nM)	[14]
τ_0	0	μM	-
N_0	0	billions of neurons lost	-
C_0	0	MMSE test score decline	-
a_1	6.5641	$\mu\text{M} \cdot \text{year}^{-1}$	[13] adjusted
a_2	1.5778	$\mu\text{M} \cdot \text{year}^{-1}$	[21] adjusted
b_1	315569260	nM $\cdot \text{year}^{-1}$	[21] lets review this one
b_2	6311385.2	year^{-1} (100)	[21]
c_1	52.2958	$\mu\text{M} \cdot \text{year}^{-1}$	[15]
c_2	1.78367	year^{-1}	[13]
c_3	0.1	year^{-1} (1/100)	no source
d_1	0.0716	billions of neurons lost $\cdot \text{year}^{-1}$	no source
d_2	0.398406	billions of neurons lost $\cdot \text{year}^{-1} \cdot \mu\text{M}^{-1}$	[15]
e_1	0.01463	MMSE decline $\cdot \text{year}^{-1}$	no source
e_2	0.008684	MMSE decline $\cdot \text{year}^{-1} \cdot \text{billions neurons lost}^{-1}$	[6] adjusted
e_3	0.01992	MMSE decline $\cdot \text{year}^{-1} \cdot \mu\text{M}$	[6] adjusted
k_1	0.303598	year^{-1}	[13] adjusted
k_2	3155692.6	year^{-1}	[21]
k_3	1.8333	year^{-1}	no source
k_4	0.3588	year^{-1}	no source (?)
k_5	0.08684	year^{-1}	no source
u_1	varies	year^{-1}	-
u_2	varies	year^{-1}	-
u_3	varies	year^{-1}	-
σ	50 – 150	nM	no source (?)

5 Sensitivity Analysis

Sobol sensitivity analysis is used to provide a quantitative assessment of the impact model parameters have on the causal dynamics of the proposed model. This computation was performed using Python, with the SALib library [9, 8]. The first, second, and total-order indices (S_T) were computed using different sample sizes, ranging from $N = 2^4$ to $N = 2^{16}$. Using the parameter ranges from Table 1, the first-order and total-order sensitivity indices are highlighted in Figure 1; initial conditions were set for $Ca(0) = 100$, and zero for the rest of the state variables. In order to validate these indices, Figure 2 shows the six most sensitive parameters for each sample size. The total-order indices are quantified along with the confidence interval for each parameter at $N = 2^{16}$ in Table 2. Using a threshold of 10^{-4} , the parameters falling below this threshold provide a negligible impact on the overall model. From this, it is observed that the growth of calcium in Equation (3) is suggested to have very little effect on tau protein synthesis, which corroborates with findings in some studies (list citations).

Table 2: Total Sobol Sensitivity indices for various parameters with corresponding confidence intervals for $N = 2^{16}$

Param	k5	k3	e3	k1	c1	c2	a1
ST	2.791e-01	2.756e-01	1.224e-01	5.973e-02	5.505e-02	4.745e-02	3.822e-02
Conf Int	3.103e-03	3.108e-03	1.441e-03	6.114e-04	6.182e-04	5.667e-04	4.264e-04

Param	k4	e2	r	d2	k2	c3	a2
ST	3.051e-02	2.995e-02	2.992e-02	2.978e-02	5.006e-03	4.915e-03	2.209e-03
Conf Int	3.358e-04	3.693e-04	3.630e-04	3.453e-04	6.306e-05	5.278e-05	2.552e-05

Param	b1	b2	e1	d1	sig
ST	2.076e-03	6.138e-04	1.602e-05	2.445e-07	1.865e-14
Conf Int	2.592e-05	6.276e-06	1.866e-07	3.464e-09	2.977e-14

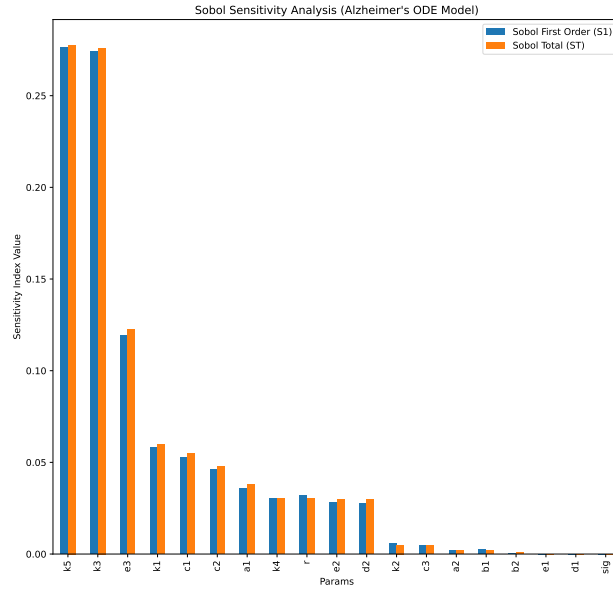


Figure 1: First and Total order indices at $N = 2^{10}$

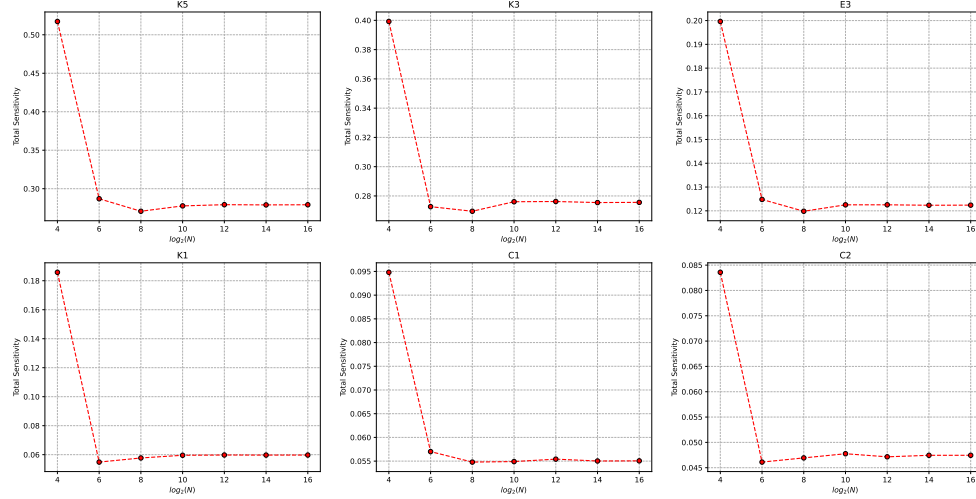


Figure 2: 6 most sensitive parameters for $N \in [2^4, 2^{16}]$

6 Results

Using the derived biological parameters in Table 1, we run the simulation forward for 50 years with the initial conditions $(A_0, Ca_0, \tau_0, N_0, C_0) = (0, 100, 0, 0, 0)$. This simulation uses `solve_ivp` from the SciPy package in Python. The results from the simulation are shown in Figure 3. Initially, A_β and Ca increase and drive each other's growth. This leads to a large growth of τ proteins, thus leading to neuron loss over time, and cognitive decline over an even longer time scale. The biomarker populations reach their equilibrium values relatively early, but it takes cognitive decline many years to reach its equilibrium value.

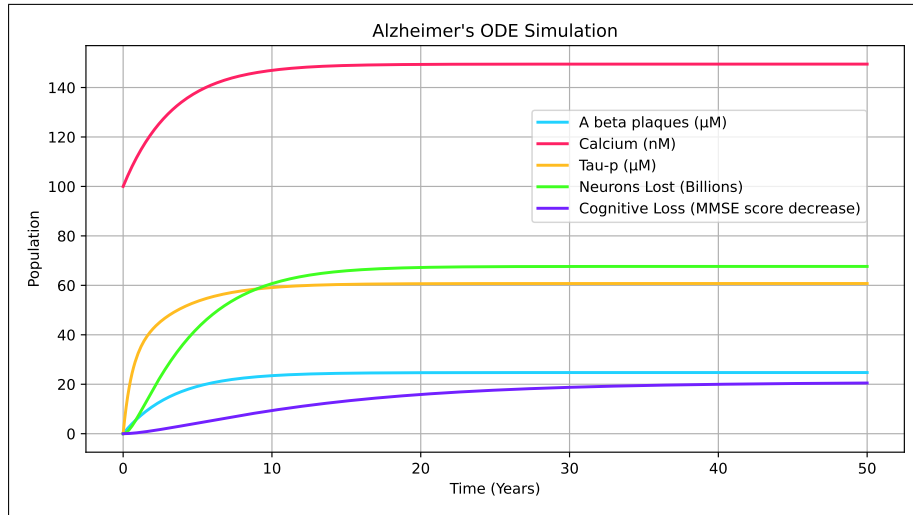


Figure 3: ODE system simulation

A phase portrait (Figure 4) shows how different trajectories from varying initial conditions traverse the $A_\beta - Ca$ space. Trajectories will primarily move toward the Ca nullcline with slight movement along the horizontal axis. This is because the time-scale of Ca is much

smaller than that of A_β . Once trajectories are near this nullcline, they travel directly toward the equilibrium solution.

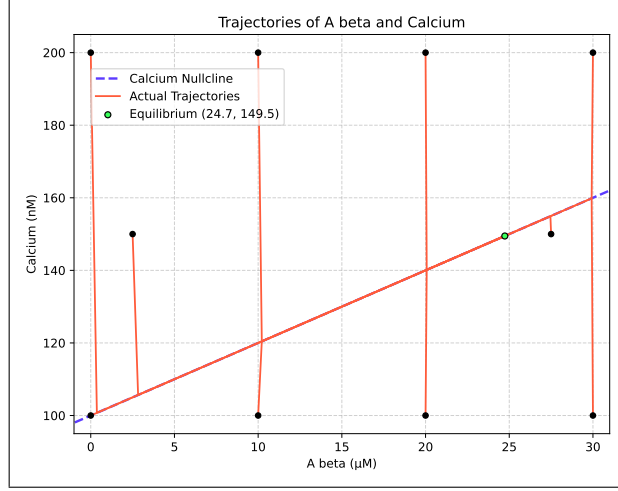


Figure 4: Phase portrait

To determine which proteins are most effective to treat, we graphed the equilibrium cognitive decline value against varying treatments against each protein, as seen in Figure 5. Each u_n term increases such that the behavior of the function can be observed. We see that the variable that has the largest impact on C^* is u_3 , the treatment that reduces the number of τ proteins. The next most effective treatment is u_1 , which reduces the concentration of A_β plaques in the brain. We observe that treating the average level of Ca does not have a large effect on the equilibrium cognitive decline.

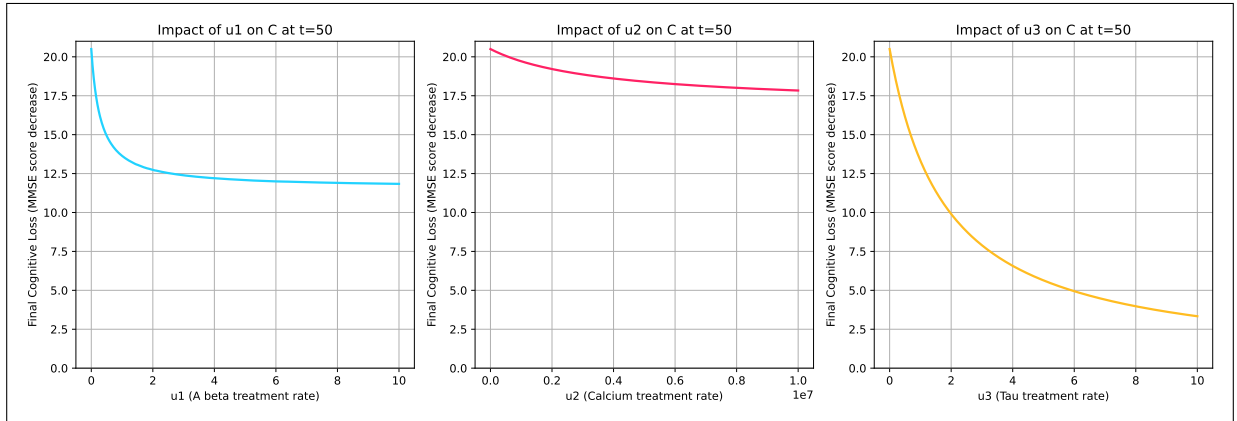


Figure 5: Treatment effect on C^*

7 Optimal Control

Optimal control is a method we can use to determine the optimal values of u_n while tracking some cost function. In optimal control, we will let each u_n be a function of time, $u_n(t)$,

as it is not realistic to maintain a constant treatment over a long time-frame. Tracking a cost function is useful for our application as some drugs used to treat Alzheimer's Disease have been shown to induce strong side effects such as brain swelling and brain bleeding (citation needed). So, minimizing this cost function while maximizing treatments will let the medication have its strongest impact without crossing some side-effect threshold.

7.1 Objective Function

The following equation is our objective function which tracks our cost over the time horizon of the treatment.

$$J(u_1, u_2, u_3) = \int_0^K A_1 N + A_2 C + A_3 u_1^2 + A_4 u_2^2 + A_5 u_3^2 dt. \quad (8)$$

The goal of treatment is to minimize the equilibrium values N^* and C^* while at the same time minimizing the side effects from the drugs. The terms A_n are the weights which determine how important each associated variable is in the optimization problem. The variables u_n are the treatment inputs, subject to the following constraints:

$$u = \{(u_1, u_2, u_3) \in L^\infty(0, K)^3 \mid (u_1, u_2, u_3) \in [0, u_1^{\max}], [0, u_2^{\max}], [0, u_3^{\max}] \forall t \in [0, K]\}$$

Each term u_n must be non-negative during the duration of the treatment.

7.2 Hamiltonian

$$\begin{aligned} \mathcal{H} = & A_1 N + A_2 C + A_3 u_1^2 + A_4 u_2^2 + A_5 u_3^2 + \lambda_1 \left(a_1 + a_2 \frac{Ca}{Ca + \sigma} - k_1 A_\beta - u_1 A_\beta \right) \\ & + \lambda_2 (b_1 + b_2 A_\beta - k_2 Ca - u_2 Ca) + \lambda_3 (c_1 + c_2 A_\beta + c_3 Ca - k_3 \tau - u_3 \tau) \\ & + \lambda_4 (d_1 + d_2 \tau - k_4 N) + \lambda_5 (e_1 + e_2 NR + e_3 \tau - k_5 C). \end{aligned}$$

The Hamiltonian includes the cost function as described in Equation (8) as well as the co-state vectors for each variable present in our system. Thus, we seek an optimal control set (u_1^*, u_2^*, u_3^*) such that

$$J(u_1^*, u_2^*, u_3^*) = \min \{J(u_1, u_2, u_3) \mid (u_1, u_2, u_3) \in \mathcal{U}\}.$$

more stuff to add here

Theorem 2 (Existence of Optimal Control). *There exists an optimal control set (u_1^*, u_2^*, u_3^*) that minimizes $J(u_1, u_2, u_3)$ over $\mathcal{U}$. Furthermore, there exists adjoint variables $\lambda_1, \lambda_2, \lambda_3, \lambda_4$ and λ_5 that satisfy the following system of differential equations:*

$$\begin{cases} \frac{d\lambda_1}{dt} = \frac{-\partial \mathcal{H}}{\partial A_\beta} = k_1\lambda_1 + u_1\lambda_1 - b_2\lambda_2 - c_2\lambda_3 \\ \frac{d\lambda_2}{dt} = \frac{-\partial \mathcal{H}}{\partial Ca} = -\lambda_1 \frac{a_2\sigma}{(Ca+\sigma)^2} + k_2\lambda_2 + u_2\lambda_2 - c_3\lambda_3 \\ \frac{d\lambda_3}{dt} = \frac{-\partial \mathcal{H}}{\partial \tau} = k_3\lambda_3 + u_3\lambda_3 - d_2\lambda_4 - e_3\lambda_5 \\ \frac{d\lambda_4}{dt} = \frac{-\partial \mathcal{H}}{\partial N} = -A_1 + k_4\lambda_4 - e_2R\lambda_5 \\ \frac{d\lambda_5}{dt} = \frac{-\partial \mathcal{H}}{\partial C} = -A_2 + k_5\lambda_5 \end{cases}$$

with transversality conditions $\lambda_1(K) = \lambda_2(K) = \lambda_3(K) = \lambda_4(K) = \lambda_5(K) = 0$. Also, the optimal control set has the characterization:

$$\begin{cases} \frac{\partial \mathcal{H}}{\partial u_1} = 2A_3u_1 - \lambda_1A_\beta = 0, & u_1^* = \min \left\{ u_1^{\max}, \max \left\{ 0, \frac{\lambda_1A_\beta}{2A_3} \right\} \right\} \\ \frac{\partial \mathcal{H}}{\partial u_2} = 2A_4u_2 - \lambda_2Ca = 0, & u_2^* = \min \left\{ u_2^{\max}, \max \left\{ 0, \frac{\lambda_2Ca}{2A_4} \right\} \right\} \\ \frac{\partial \mathcal{H}}{\partial u_3} = 2A_5u_3 - \lambda_3\tau = 0, & u_3^* = \min \left\{ u_3^{\max}, \max \left\{ 0, \frac{\lambda_3\tau}{2A_5} \right\} \right\}. \end{cases}$$

Proof. The □

7.3 Numerical Results

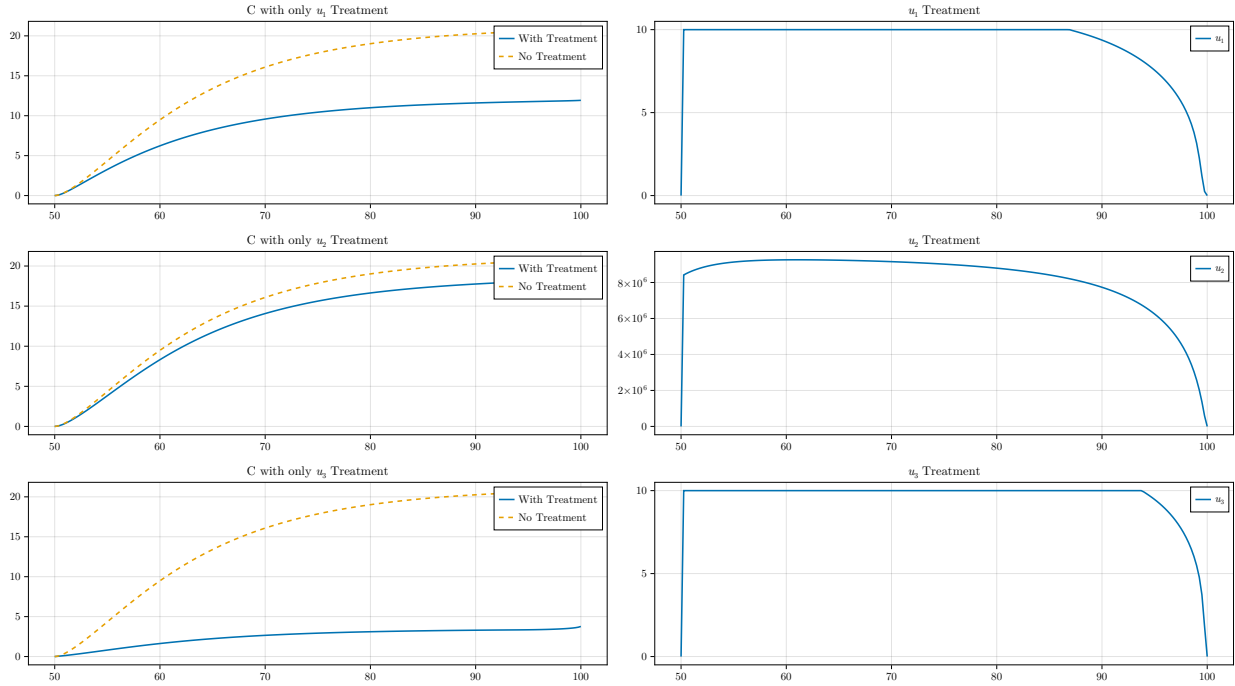


Figure 6: Optimal control simulation results

- explain how weights were calculated

After running numerical simulations for optimal control, we find that the results are in line with the simulation shown in Figure 5. Treatments that target τ -protein are most

effective in reducing C^* , followed by treatments targeting A_β . Again, treatments targeting Ca are found to be the least effective.

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